IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- This assignment involved retrieving data, preparing data, summarizing and visualizing data, and evaluating whether variables retrieved offer a means to determine whether the stage one boosters of Falcon 9 launches would successfully land upon the landing pads.
- Some findings of note:
  - Visually, flight number/year of the launch showed increases in success over the timeframe of 2010 to 2020, Space X presumably improved the successfulness of the landings over this timeframe. The most "accurate" predictive model (a Decision Tree Classifier) found that the flight number was the second highest variable by importance
  - The most important variable was absence of gridfins.
  - Other variables notably were the landing pad, the launch site, the number of flights that the stage 1 previously flew, and the orbit.
  - Visually, a low or extreme payload appeared to have some affect on the success of the landing; however, this may be accounted for in the model by the flight number, grid fins, or other measure.

Introduction

Project background and context

- Firing rockets into space to deploy satellites is a prohibitively expensive venture. Space X's Falcon 9 rocket offers a more competitive cost (62 million per launch) relative to its competitors (165 million dollars per launch). One reason Space X offers reduced costs is because the company reuses a portion of its rocket, should it land successfully, reducing overhead.
- Data exist on the outcomes (success and failure) of Space X Falcon 9 launches.

Project Goal

- The aim of this project was to analyze the data that exists.
  - To offer summaries and visualizations that aid in understanding of variables that were collected on the launches, and
  - To prepare a model where its independent variables inform of successful and failed recoveries of the first stage of Space X's Falcon 9 Rocket for single payload launches.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - Code collected data through the scraping websites or reading in files from Space X, Wikipedia, and IBM.
- Perform data wrangling
  - The Pandas library in Python aided in preparation of the data files analyzed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Predictive analysis on the successful landing was run using four different models. A Random Forest Predictive Model was the most successful model with an > 80% “predictive accuracy”.

Data Collection

How the data were collected.

- The following Python libraries permitted extraction and preparation of the data files: *requests, pandas, numpy, datetime*.
- Functions with REST API calls extracted the information for several variables including information on the rocket, launchpad, payloads, and specific data on the number of cores, reuse of the core, landing success from the launch, flight, presence of gridfins,
- From this data, a new dataset
    - combined data extracted via the functions,
    - subset the extracted data so that the new dataset only consisted relevant variables,
    - Limited the data to Falcon 9 launches having single cores and one payload.

Data Collection – SpaceX API

Import Libraries
- Import libraries
  - pandas, requests, numpy, and datetime

Define functions
- Define functions
  - getBoosterVersion(), getLaunchSite(), getPayloadData(), getCoreData() – inputs a data file and outputs a number of variables into separate Python lists

Read in data
- Read in source .json file
  - Obtained using requests. Pandas converts the .json file into a pandas dataframe using .json.normalize()

Data preparation
- Pandas prepares a dataframe
  - Pandas filters the data for launches with just a single core and payload. Datetime converts a date in datetime format to a date format

Run functions
- Functions are run importing the pandas dataframe outputting data into lists

Concatenate Lists
- Concatenate lists
  - A Python dictionary contaentates thte data from a list. Pandas converts the dictionary into a new dataframe.

Filter data
- Filter data
  - Pandas creates a new dataframe filtering the one of contatenated lists for only rows of the Falcon 9 rocket

Impute missing values
- Impute missing values
  - Missing values are observed in the payload field. From values present in the field, pandas calculates the mean, and adds the value to rows missing data.

My GitHub URL https://github.com/GitJubJub/IBM-Applied-Data-Science-Project/blob/main/jupyter-labs-spacex-data-collection-api.ipynb

Data Collection - Scraping

Import Libraries

- Import modules/libraries
  - sys, beautifulsoup4, bs4 from beautifulsoup4, re, unicodedata, pandas

Define functions

- Define functions
  - date_time(), booster_version(), landing_status(), get_mass(), extract_column_from_header() ,

Read in data

- Read in Wikipedia article
  - Request connects to the url. Beautifulsoup parses the html. Python finds each table in the html file, and extracts data for the 2nd table

Obtain table headers

- Extract column names from the table
  - Python loops through table header tags in the html. The extract_column_from_header() retrieves the column name. Python stores it into a Python list

Create Dataframe

- Create Dataframe
  - Python creates lists in dictionaries for variables. Python loops through rows, and appending data into lists. Pandas creates a dataframe looping through dictionaries and lists

Outputting to a CSV

- Outputing data into a comma-separated value file
  - Pandas outputs the dataframe into a comma-separated value file using the to_csv() function.

My GitHub URL https://github.com/GitJubJub/IBM-Applied-Data-Science-Project/blob/main/jupyter-labs-webscraping.ipynb

Data Wrangling

Import Libraries

- Import libraries
  - Pandas, and numpy

Read in data

- Read in data
  - Pandas reads in a comma-separated value file with the data

Check of data condition

- Check of data quality/condition
  - Pandas checks for calculates the completeness of each column. Pandas informs of the data variable type of each field.

Calculate frequencies

- Calculate the frequencies of several categorical variables
  - Pandas generates frequency counts for several categorical variables including: landing outcome, orbit, and the launch site.

Create a binary outcome measure

- Create a binary measure of the landing outcome to serve as the dependent variable for further analysis
  - Pandas creates a variable called “Class”. Numpy creates an array derived whether items in a list are defined as a unsuccesful outcome.

My GitHub URL https://github.com/GitJubJub/IBM-Applied-Data-Science-Project/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb

EDA with Data Visualization

- Categorical plots (catplot) from the seaborn library presented graphs to evaluate the relationship of the payload mass, launch site, Falcon 9 flight numbers with the landing outcome
- A bar graph (barplot) from the seaborn library showed the percent of a successful landing depending upon the orbit of the launch
- A line plot from the seaborn library illustrated the success rate of the landings across the timeframe of 2010 to 2020.

My GitHub URL https://github.com/GitJubJub/IBM-Applied-Data-Science-Project/blob/main/edadataviz.ipynb

EDA with SQL

- Sqlite ran the following queries on the data
  - Retrieved distinct launch sites from the data; and distinct landing outcomes in the data
  - Displayed the first five entries of the CCA* launch site.
  - Totaled the total payload mass where the customer was NAS (CRS); and averaged payload mass where the booster version was F9 v1.1
  - Obtained the first date where a ground landing pad was present
  - Presented distinct booster version where a successful landing outcome occurred on a drone ship and payload was between 4000 and 6000 kgs; and distinct booster versions where the payload mass was highest
  - Counted total number of mission outcomes; and ranked number of landing outcomes according to their frequency between the dates of 06/04/201 and 03/20/2017
  - Listed records for the month, landing outcome, booster version and launch site, where the drone ship landing was a failure in 2015

My GitHub URL https://github.com/GitJubJub/IBM-Applied-Data-Science-Project/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

- Summarize what map objects such as markers, circles, lines, etc. you created and added to a folium map
  - Markers and circles were added to the map for the location of each launch site.
  - Markers were noted of each launch outcome for each location: those with success were colored green, and failures colored in red.
  - A MousePosition returned the coordinates (latitude and longitude) of the mouse position on the map. The data analyst retrieved the coordinates of closest highways, coastlines, and railways, relative to the launch sites.
  - Using the mouse position coordinates, the data analyst added a line (PolyLine) from a site to the proximal highway/coastline/railway.

My GitHub URL: https://github.com/GitJubJub/IBM-Applied-Data-Science-Project/blob/main/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

- Summarize what plots/graphs and interactions you have added to a dashboard
- The dashboard consisted of two plots.
  - The first plot was a pie chart. Users can select the desired pie chart from a dropdown menu. When users select “All Sites” (also the default), a pie chart consists of a percentage from each site. Otherwise, users may select a specific launch site, the site is pie chart of launch outcomes at the site.
  - The second plot was a scatter plot of the outcomes (class) and the payload mass. The second plot also had a slider, where the user could select a range of payload mass to observe where outcomes were most frequent.

My GitHub URL: https://github.com/GitJubJub/IBM-Applied-Data-Science-Project/blob/main/spacex-dash-app.py

Predictive Analysis (Classification)

Import Libraries

- Import libraries and modules
  - Pandas, numpy, matplotlib.pyplot, seaborn, and Preprocessing, linear_model.LogisticRegression, svm.SVC, tree.DecisionTreeClassifier, and neighbors.KNeighborsClassifier from sklearn , train_test_split and GridSearchCV from sklearn.model_selection

Defined Function

- Created plot_confusion_matrix() function
  - Defined the function to evaluate the performance of trained models on the “unseen” test data.

Read In data

- Read In Source data
  - Pandas read in the comma-separated value file. The data consisted of 90 rows and 83 columns

Separate the dependent and independent variables

- Separate the dependent and independent variables, and training and testing dataset
  - Pandas pulled out the “class” column. Python assigned its contents to a numpy array named Y. Independent variables were standardized. Python assigned these to the numpy array X. Using the module train_test_split, sklearn spit the arrays into training and testing variables.

Fit Models

- Fit Logistic Regression models to the training data
  - From training data, the data analyst fit a standard logistic regression, a penalized ridge regression, and lasso-ed models trained with 10-fold cross-fold validation
- Fit Support Vector Machine Model
- Fit Decision Trees Models
  - From training data, the data analyst fit decision tree models using different methods to split branches (gini or entropy), altering depth of splits, and leaf and sample split criteria
- Fit K-Nearest Neighbor Models
  - From training data, the data analyst fit several k-nearest neighbor models varying models for different number of neighbors, and different algorithms to identify the nearest neighbors

Identifying the best models

- Identifying the best model
  - GridSerchCV identified the best fitting model of each model type. And a classification matrix, presented a table of predicted vs. actual outcomes. Fitting the best model of each type to the testing data, the data analyst identified the model that best identified (the highest predictive accuracy) the dependent variable.

My GitHub URL https://github.com/GitJubJub/IBM-Applied-Data-Science-Project/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

Section 2

Insights drawn from EDA

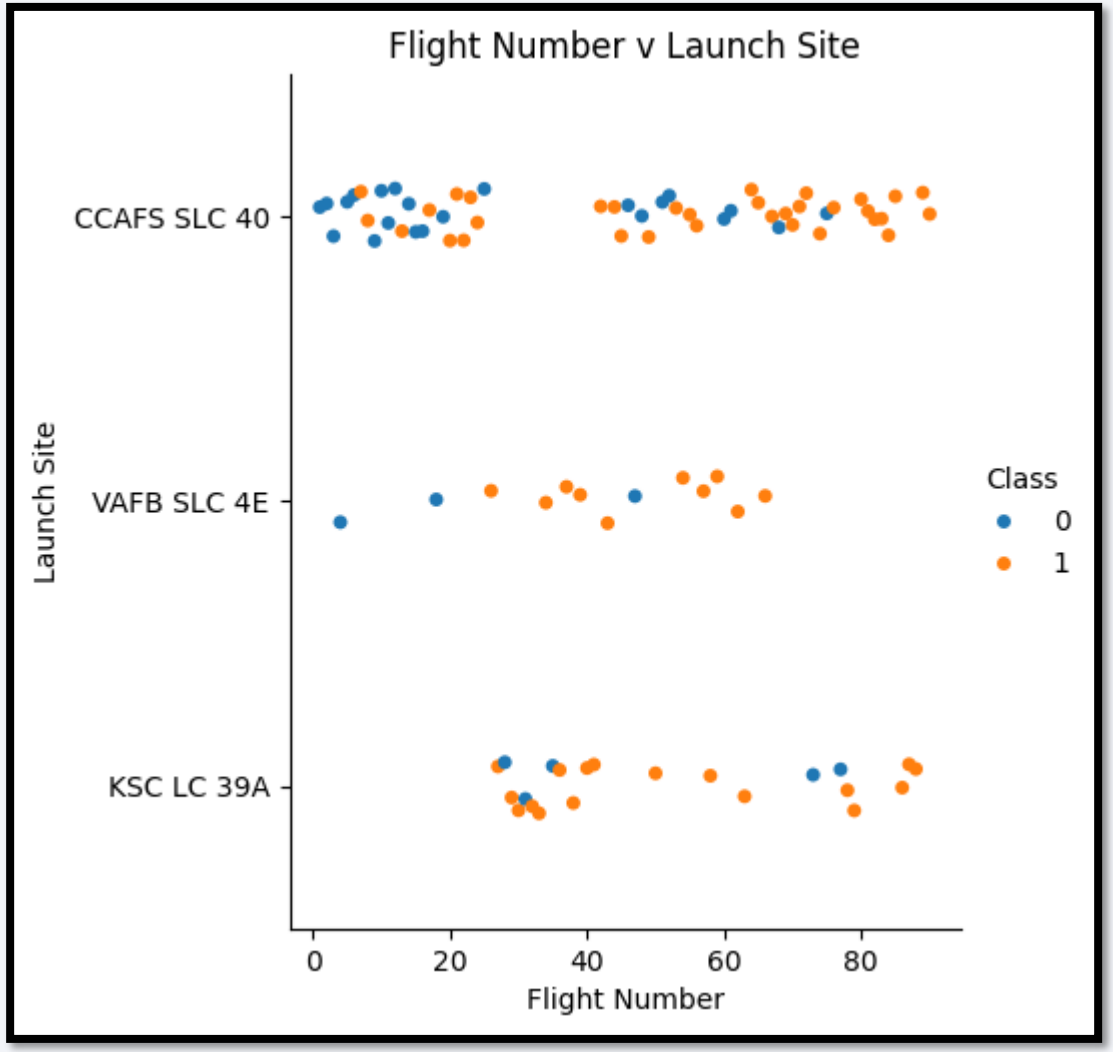
Flight Number vs. Launch Site

Observable findings:

- A noticeable gap is observed in flights from CCAFS, the launch site which also appears to be the most used site

- Flights appear to cease from the Air Force Base after flight 70.

- Success rate (class) improves across flights.

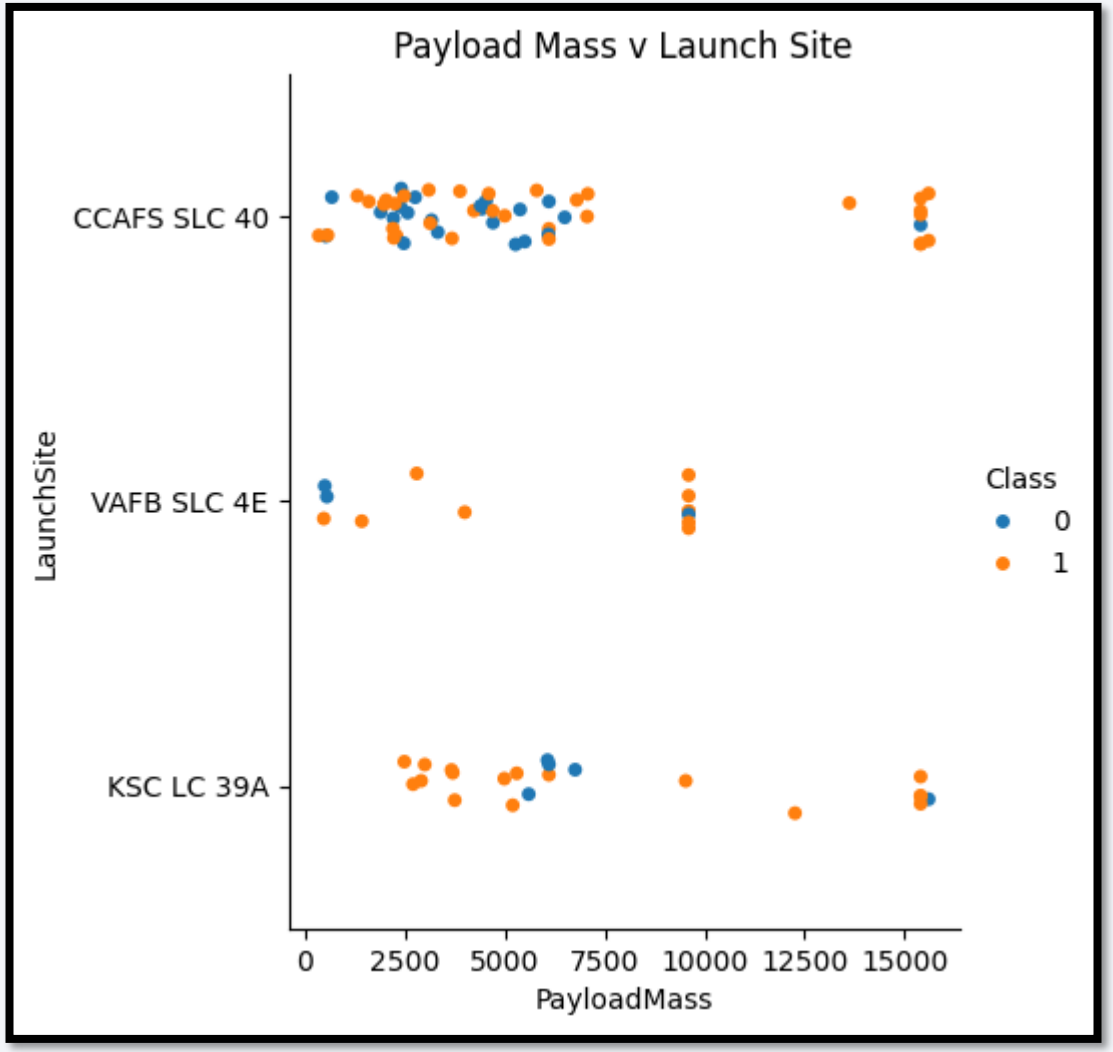


Payload vs. Launch Site

Observable Findings:

- CCAFS has gap in payload mass between 7,500 and 12,500, these appear to be launches that occurred at VAFB.

- Lower payload mass has more frequent launches thatn higher payloads.

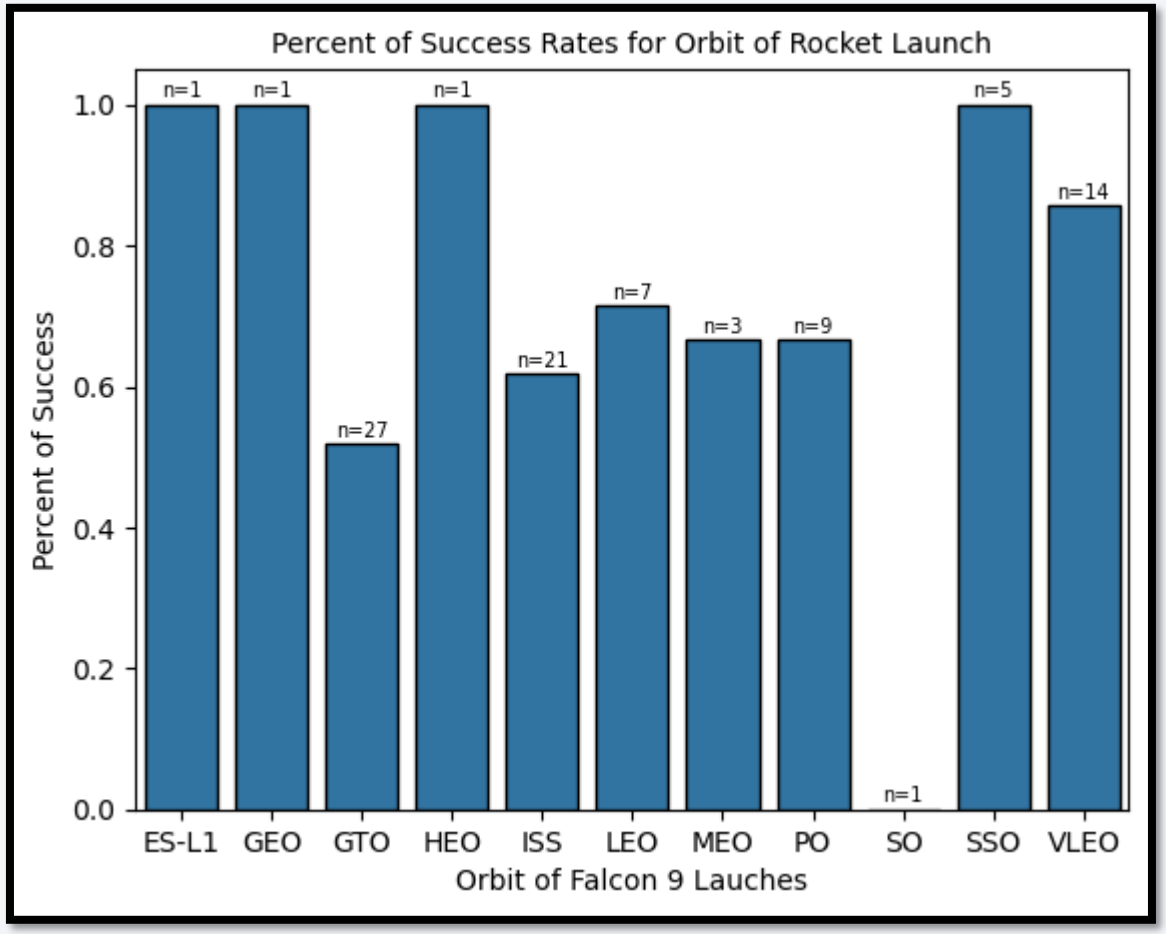


Success Rate vs. Orbit Type

Observable Findings:
- Number of payloads delivered to each orbit is not equal (ranging from 1 – 27).

- Orbit of payload may effect whether the rocket could be recovered, but other variables may be at play.

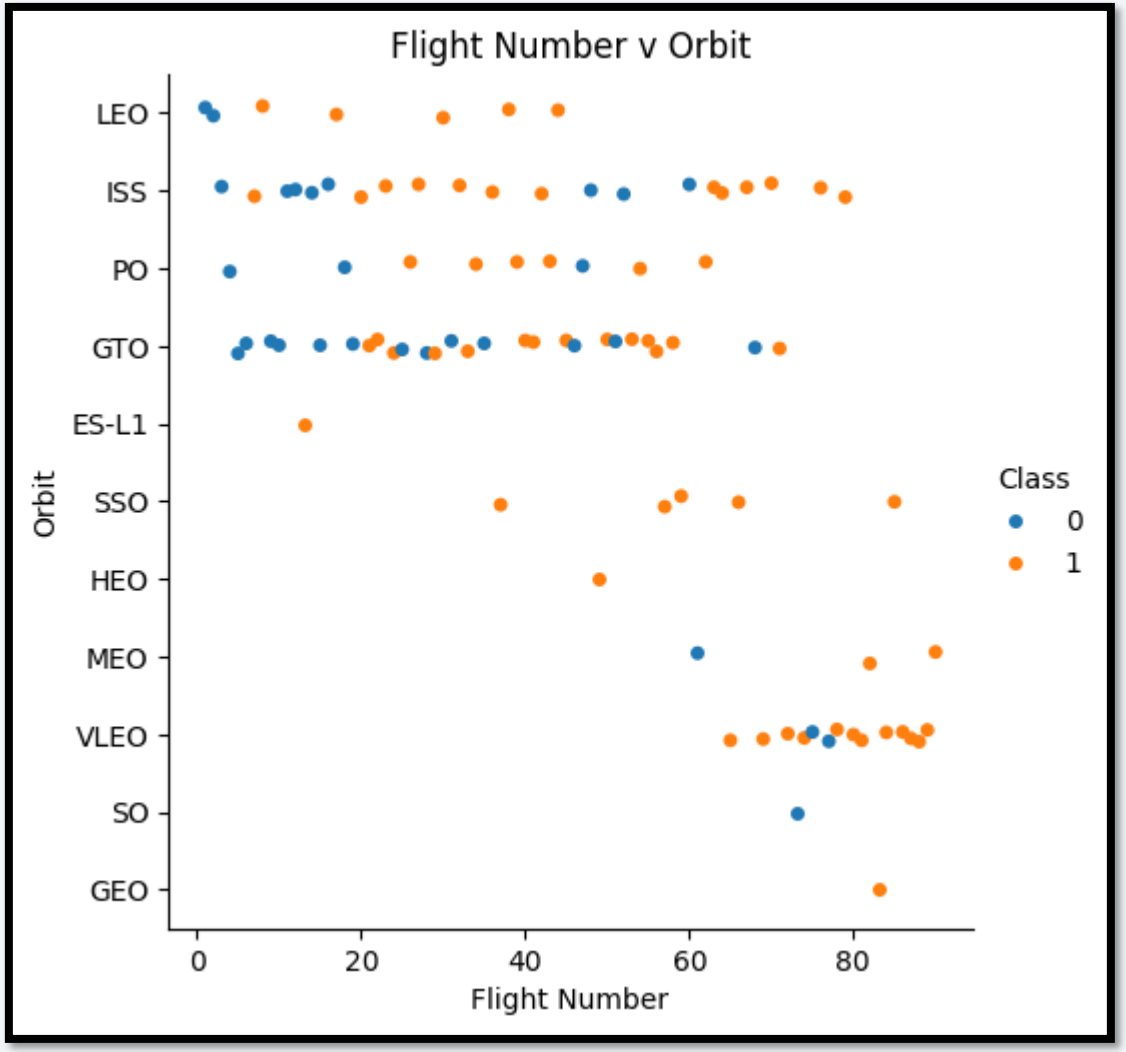
- No orbit had lower than about 50% success rate in rocket recovery. Thought some were as high as 100%.



Flight Number vs. Orbit Type

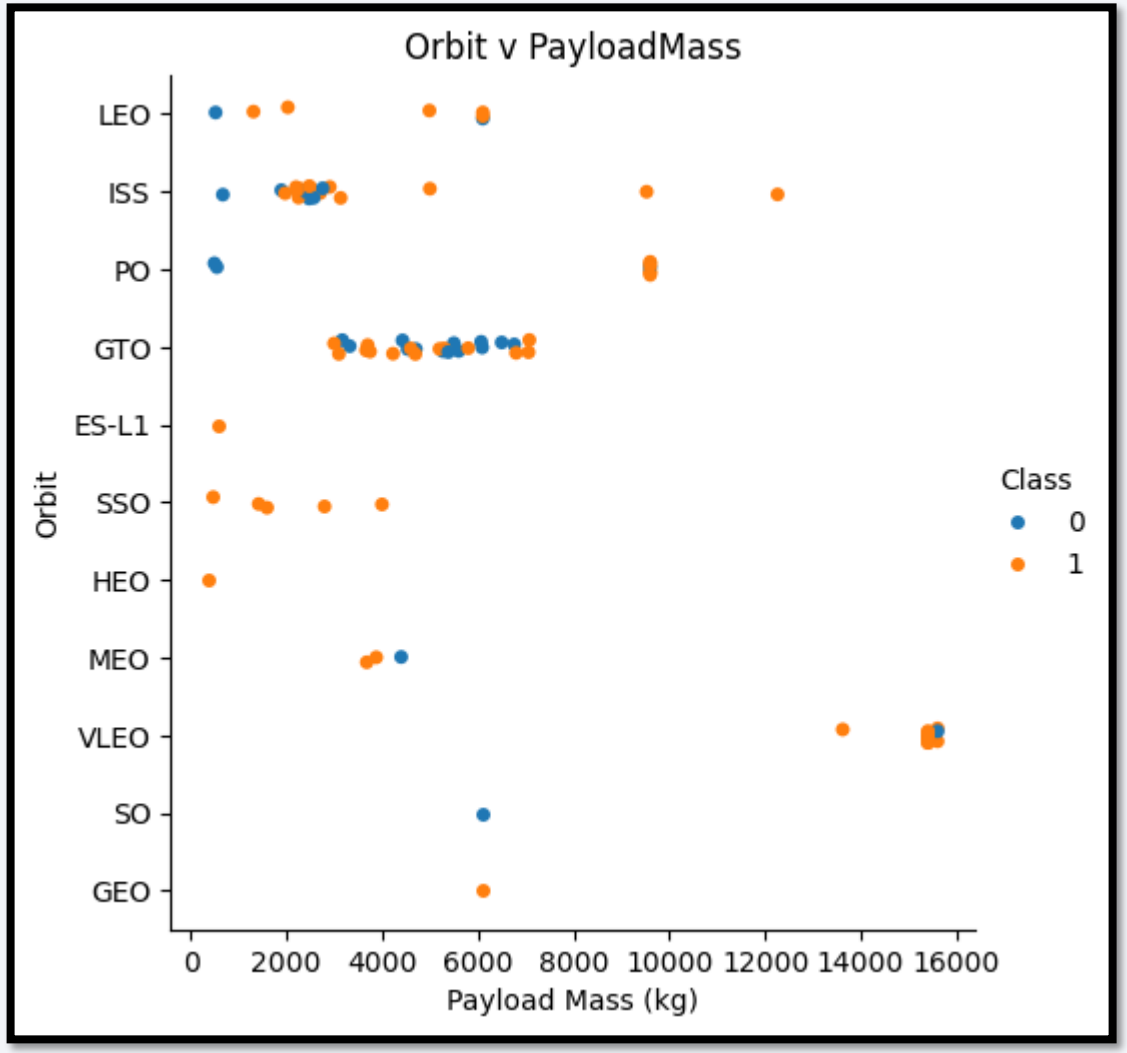
Observable Findings

- Earlier flight numbers did not have the success rate as later ones.
- Success rate might be orbit and flight number dependent
- Orbit of flight numbers appears to change over time, where most later flight numbers pertained to very low Earth orbit.



Payload vs. Orbit Type

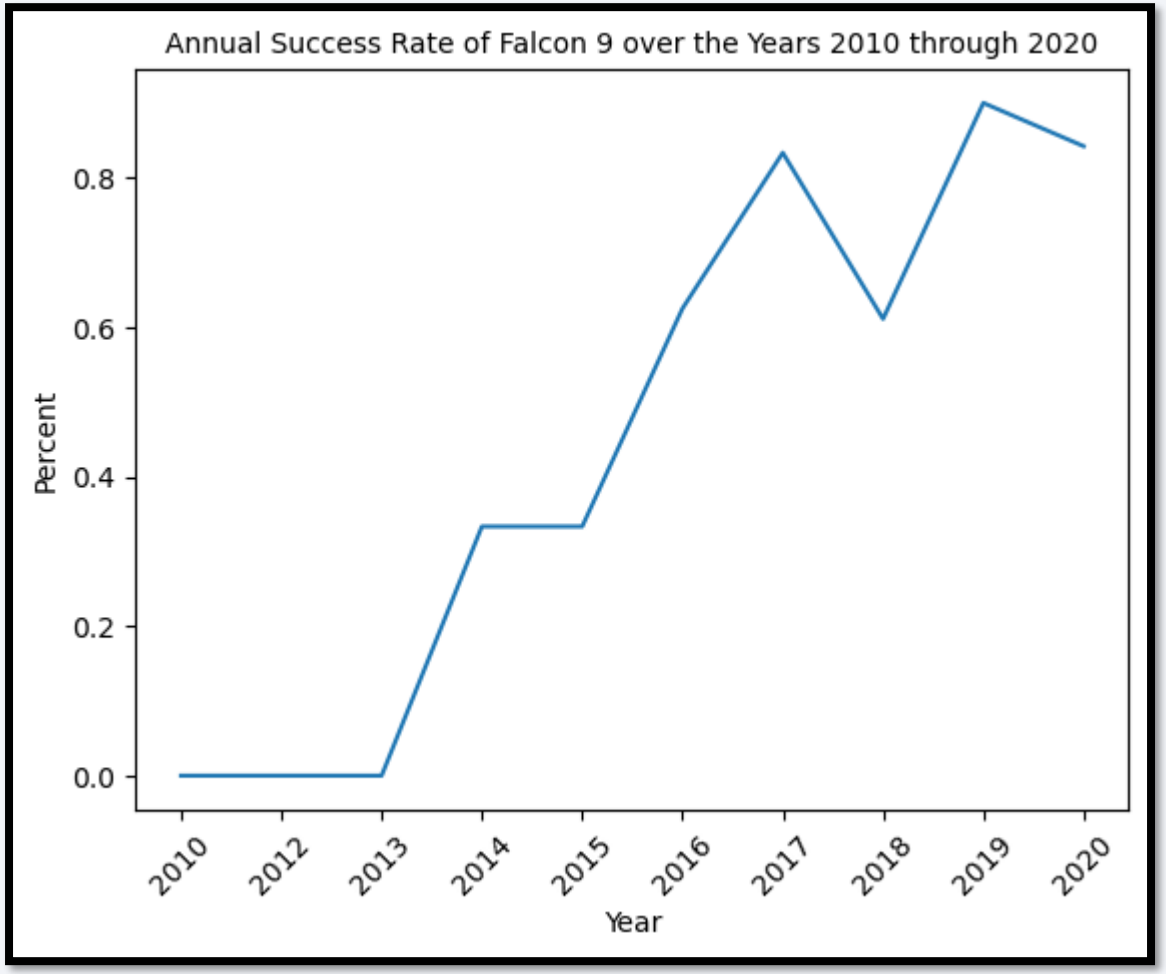
- Payload Mass and Orbit appear to show some degree of congruence, with what appear to be ranges of payloads for different orbits
- Very low Earth orbit (VLEO) launches consistently have the highest payload mass
- Geostationary transfer orbit has a distinct range of payload between 2,000 and 8,000 kgs of mass, with mixed success rates
- The lowest payload appears to have low success rate, but perhaps payload for these was low in earlier flights



Launch Success Yearly Trend

Observable findings

- Success rate correlates highly and positively with the year
- 2014 through 2017 showed the steepest increases



All Launch Site Names

- The dataset has 4 launch sites.
- CCAFS has two(?) launch sites.

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

First 5 Launch Sites Begin with `CCA`

- The data may be sorted by arranged according to data. All orbits are LEO. The variable names are present atop the table

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

Calculate the total payload carried by boosters from NASA

sum(PAYLOAD_MASS__KG_)
45596

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1

avg(PAYLOAD_MASS__KG_)
2928.4

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad

min(Date)
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

Booster_Version
F9 FT B1021.1
F9 FT B1022
F9 FT B1023.1
F9 FT B1026
F9 FT B1029.1
F9 FT B1021.2
F9 FT B1029.2
F9 FT B1036.1
F9 FT B1038.1
F9 B4 B1041.1
F9 FT B1031.2
F9 B4 B1042.1
F9 B4 B1045.1
F9 B5 B1046.1

Total Number of Successful and Failure Mission Outcomes

The total number of successful and failure mission outcomes

Mission_Outcome	count(Mission_Outcome)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

The failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

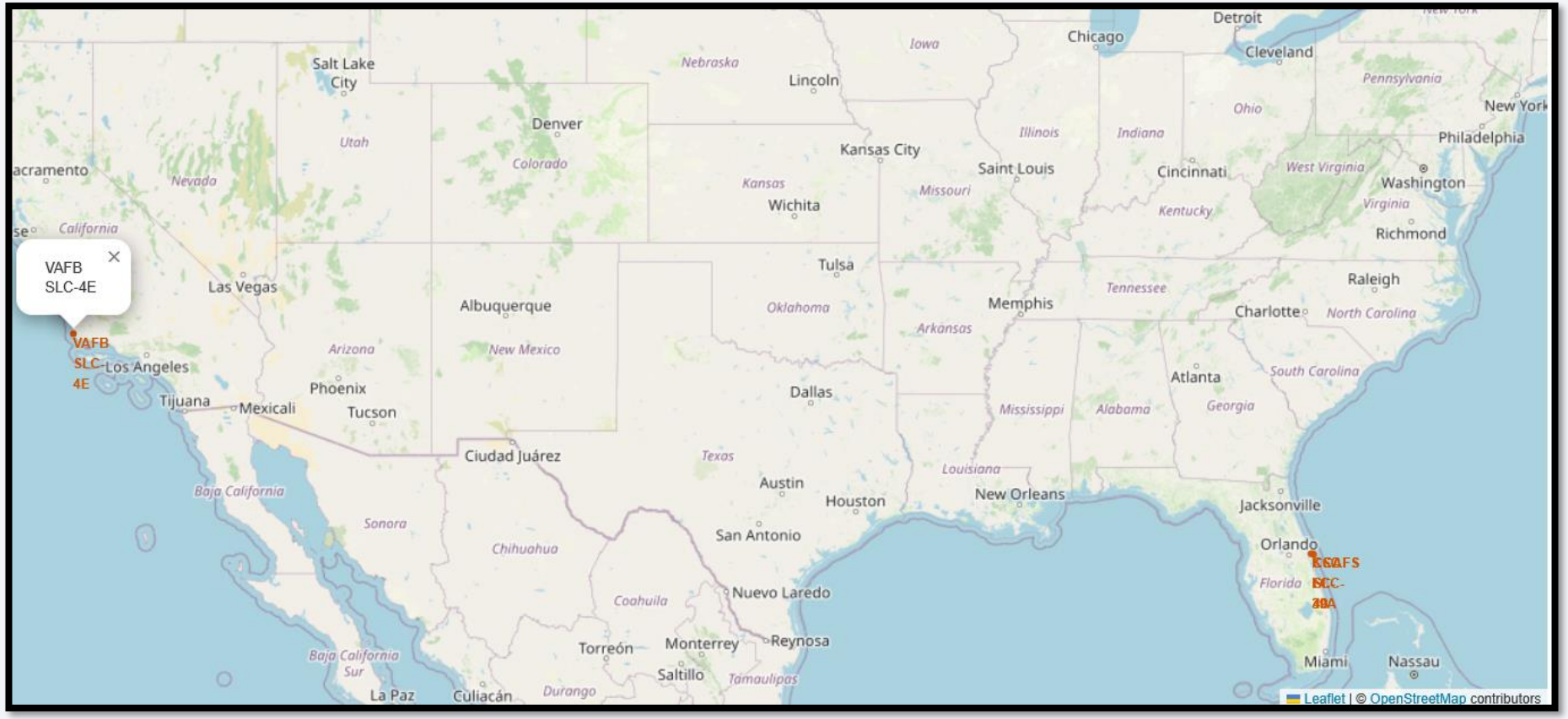
Landing_Outcome	Num_of_Rows
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Section 3

Launch Sites Proximities Analysis

Map with All Falcon 9 Launch Sites

- Vandenberg Air Force Base is located on the coast of California, northwest of Los Angeles by about 2.5 hours(?)
- The remaining launch sites (Kennedy Space Center and Cape Canaveral Air/Space Flight Station are in Florida, and very close in proximity (Maybe an hour or 1.5 hours east of Orlando)

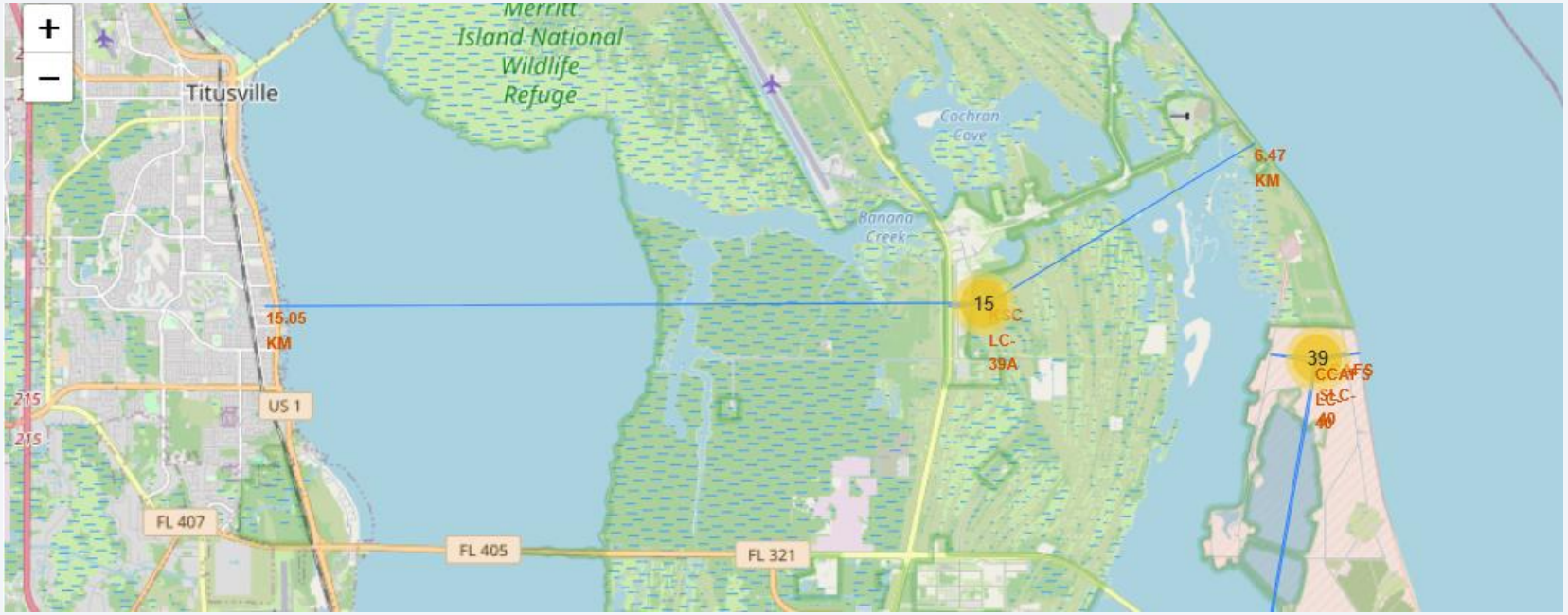


Succes and Failed Launches for each site map

- VAFB had 10 launches, 4 of success and 6 failures (see below). All other launches were in Florida.
- All launches appear close to the coasts or the edge of the US, to likely lift off over the ocean.

Distances from Launch site to Coastlines and Railways

This map shows example of the distances between KSC (Kennedy Space Center) and the most proximal residential center (Titusville) and the coastline.



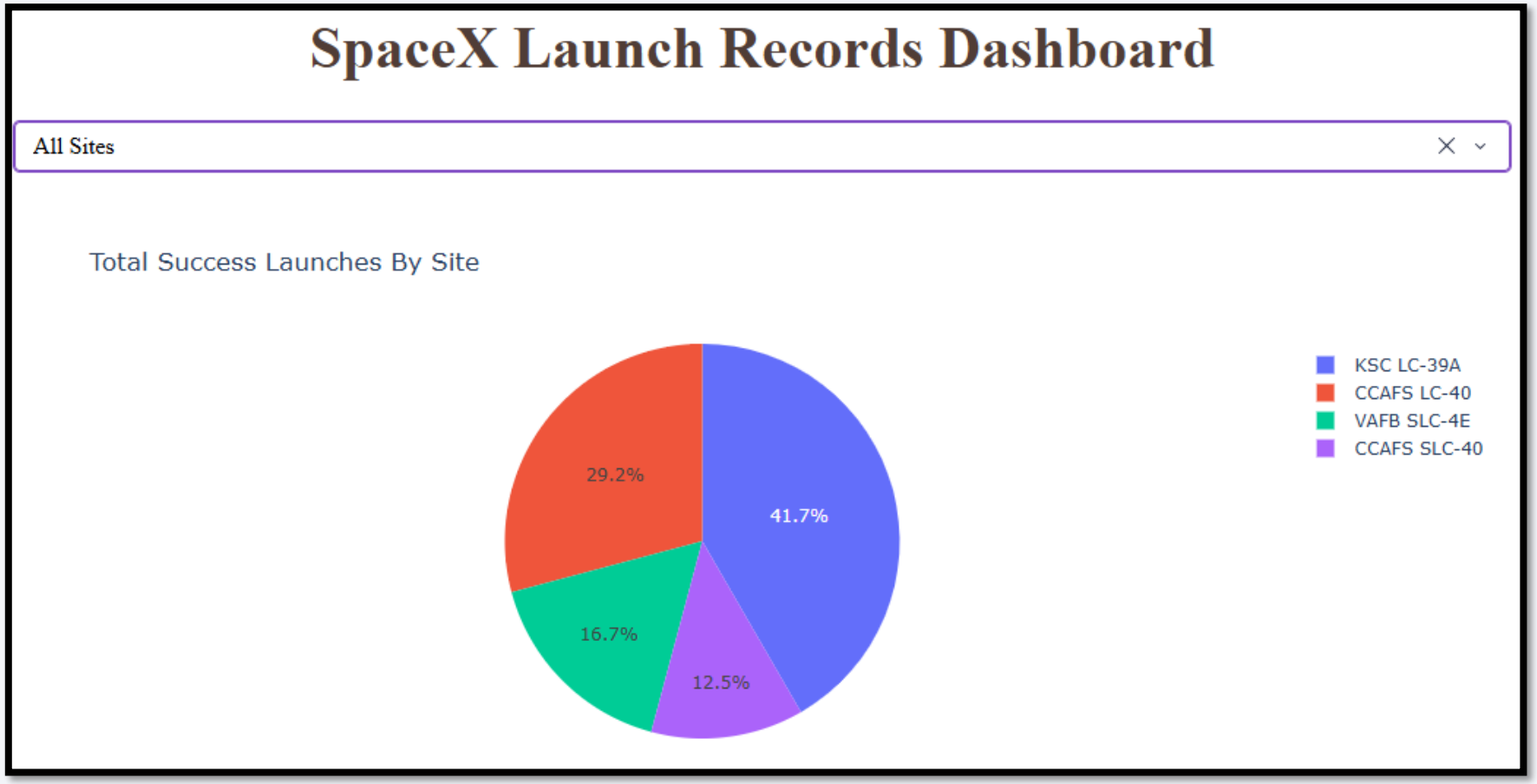
Most launchsites were close to the coast and several miles from the residential areas. Rail lines relative close to KSC, perhaps to transfer heavy items to the rocket.

Section 4

Build a Dashboard with Plotly Dash

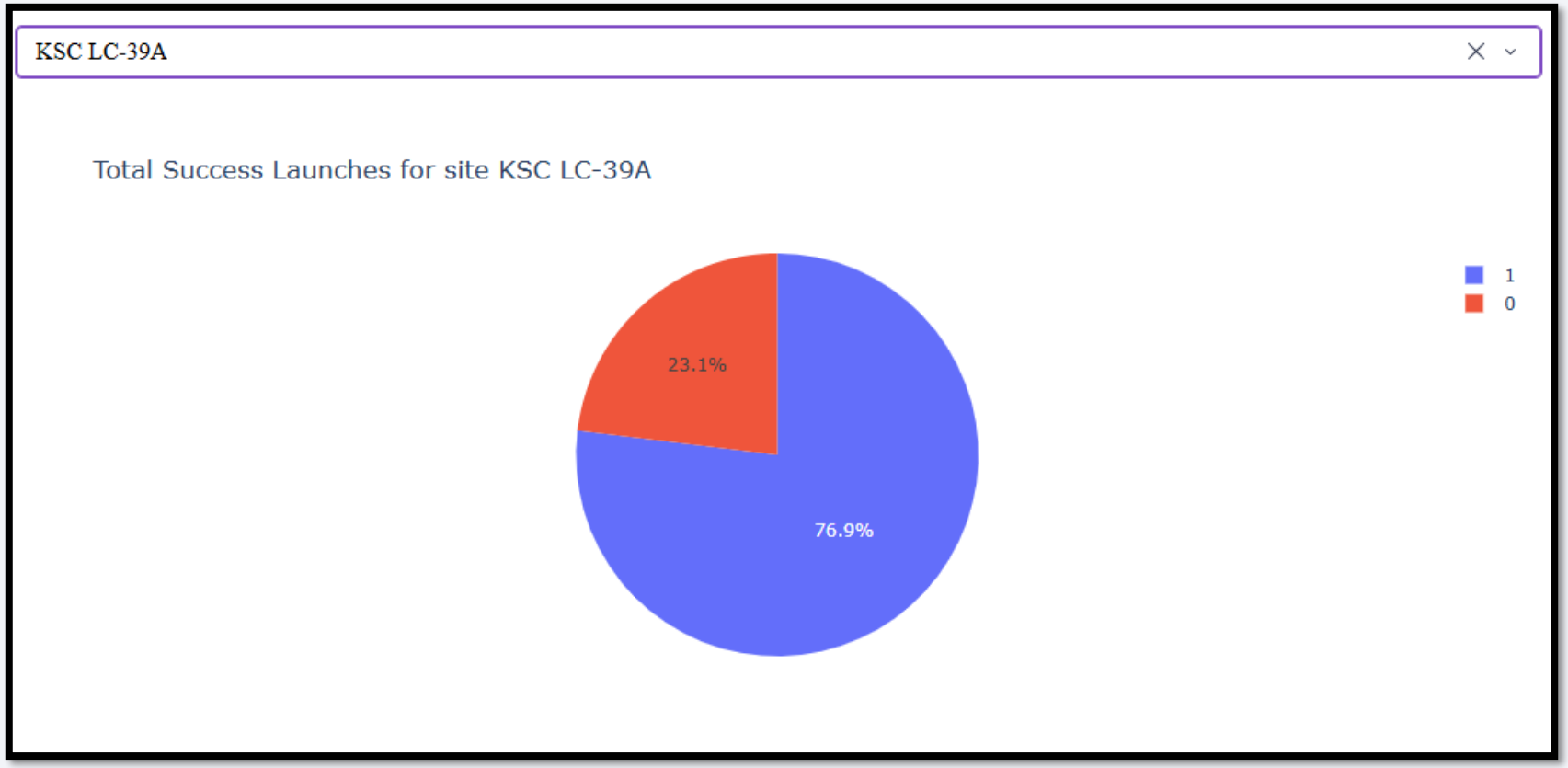
Space X Launch Records Dashboard

- Launch pads in Florida made up more than 80% of Space X launches with successful landings. The launch pads are close together.



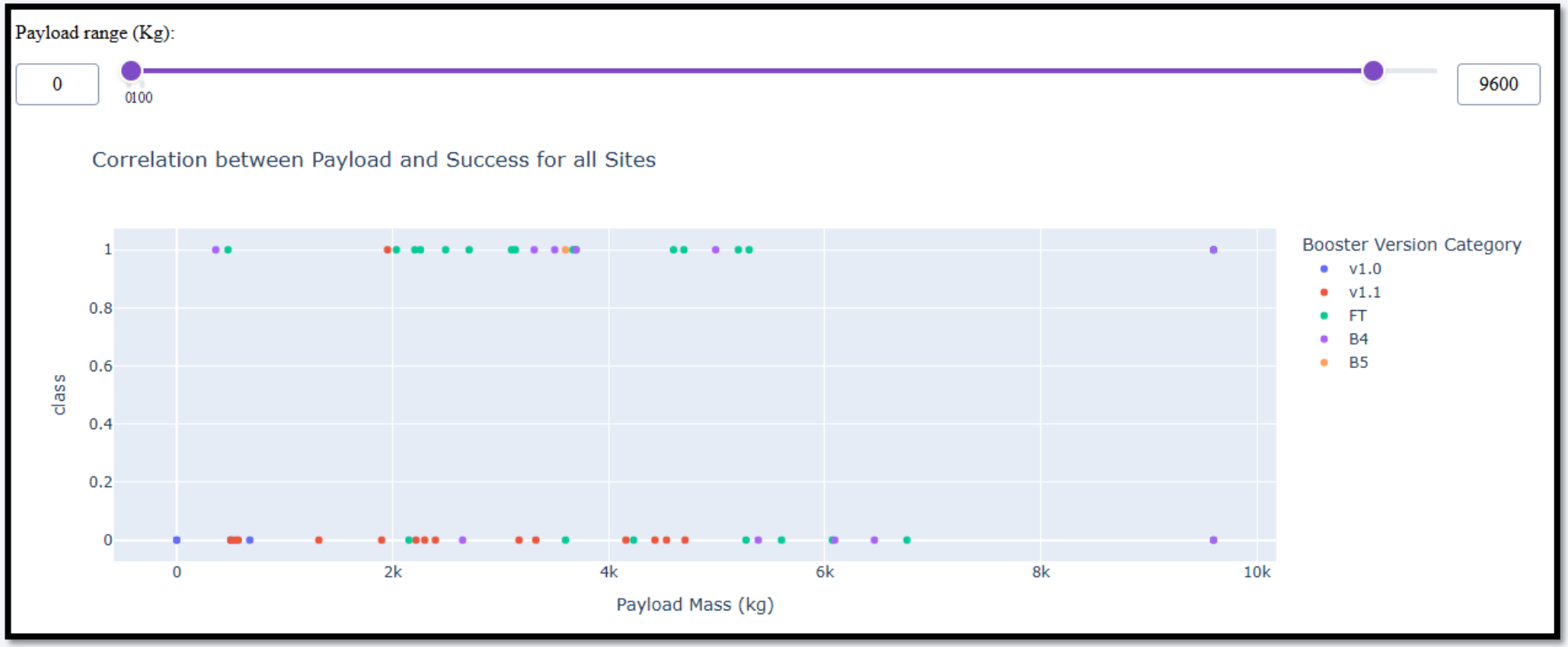
Total Successes at KSC LC-39A

- KSC LC-39A had a success rate of 76.9, compared to the others launch pads (CCAFS LC-40 was 73.1%, VAFB SLC-4E was 60%, CCAFS SLC-40 was 57.1%).



Scatter Plot of Payload v. Class (Outcome)

- Payload between 2,000 and 4,000 kgs had the highest landing success rates. Those in extremes appeared to have lower success rates.

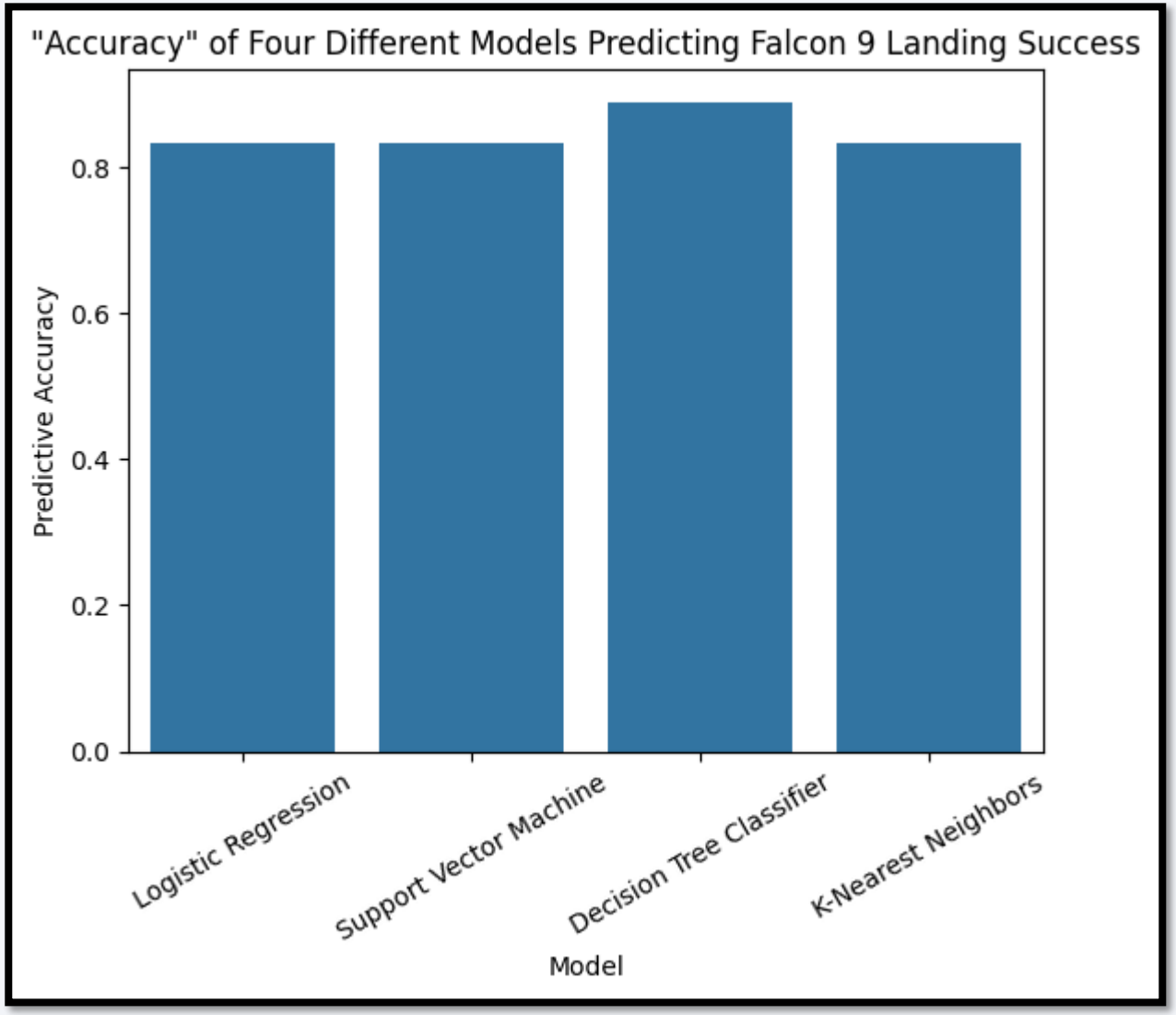


Section 5

Predictive Analysis (Classification)

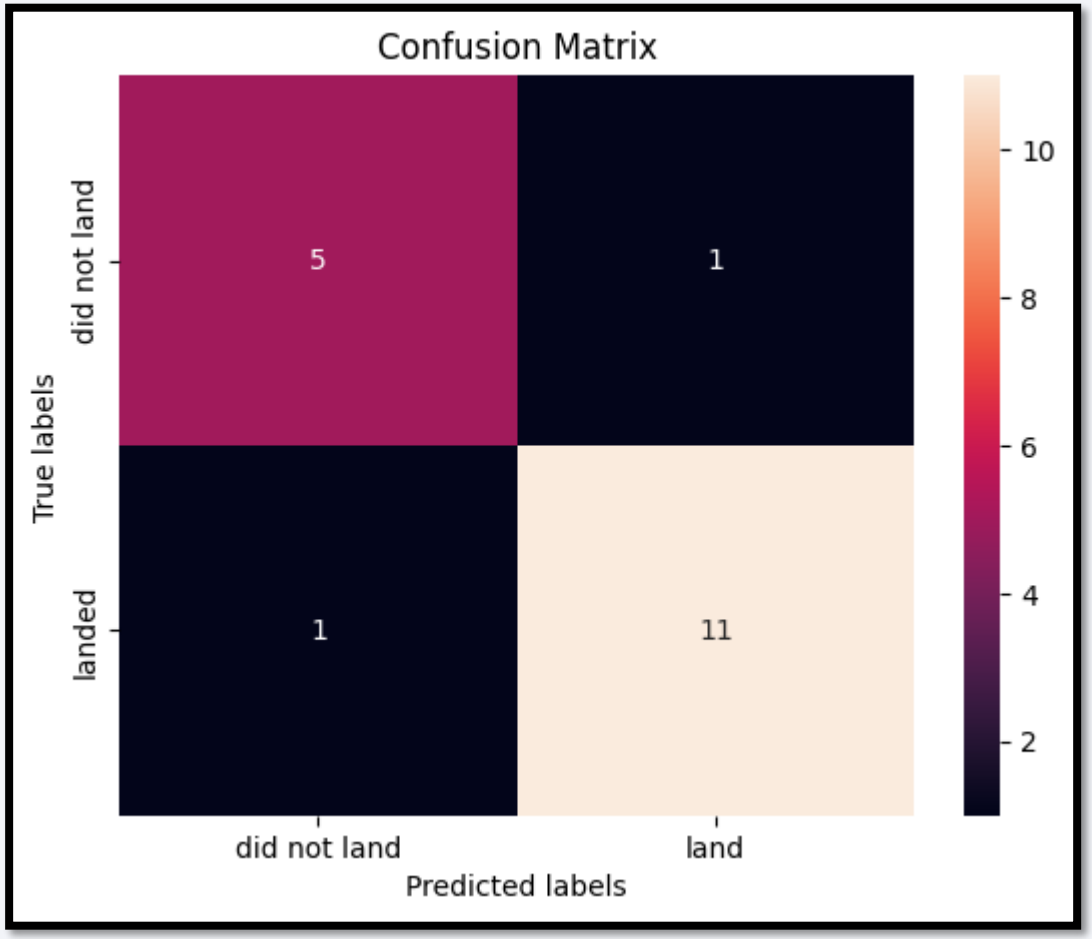
Classification Accuracy

- Accuracy was similar across the models.
- Results from the Decision Tree Classifier and the K-nearest Neighbor model seemed to vary widely depending upon the random seed used, even with identical parameters set. Then, the "champion" model may be a somewhat random selection. Especially given the training and test data size.
- Even so, all models performed well, typically above 75%, and in this case above 83 – 89%

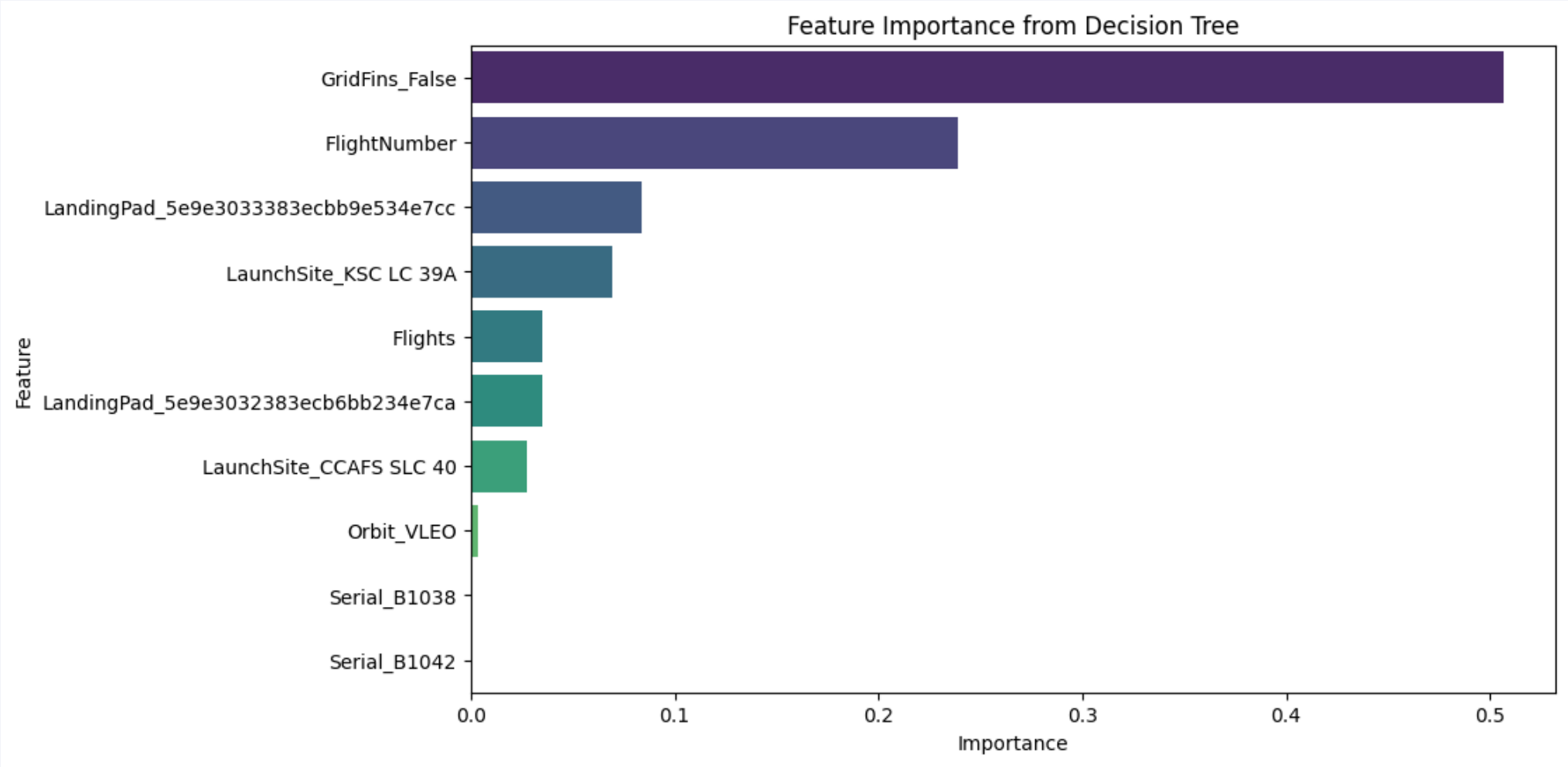


Confusion Matrix

- Confusion matrix from the highest performing model: a Decision Tree Classifier.
- The accuracy from the model in test data was 89% (*n* = 18).



Variable with the Highest Importance in the Decision Tree Classifier



Conclusions

- Over time, Space X appears to have improved the landing success of the stage 1 booster for the Falcon 9 rocket. Flight number/time was the variable with the second highest importance in a Decision Tree Classifier
- Four different predictive models offered similar outcomes. The similar outcome may be due to the small size of the training and testing dataset. In this analysis, the Decision Tree Classifier slightly outperformed the other models. The flexibility of the Decision Tree Classifier may be both a blessing (high accuracy) and a curse (potential to overfit).
- The absence of gridfins appeared to be the variable with the highest predictive importance, and landing pad was the third highest variable. Neither of these variables, the data analyst plotted in exploratory analysis.
- Conversely, launch site, payload, and orbit were variables that both appeared to have an effect on the success of the landing showed varying degrees of importance in the predictive model.

Appendix

- Brignole, A. (2026). IBM-Applied-Data-Science-Project. GitHub Repository. https://github.com/GitJubJub/IBM-Applied-Data-Science-Project

Thank you!